

# Lab 04 — Entra ID and RBAC

Set up new employee access and verify least-privilege permissions

A hands-on Azure training lab covering Microsoft Entra ID security groups and Azure role-based access control (RBAC). The lab modeled how a new employee might be given read-only access to a resource group through group membership, rather than through a direct, per-user role assignment.

|                |                    |
|----------------|--------------------|
| Resource group | rg-gp-access-model |
| Region         | North Europe       |
| RBAC role      | Reader             |
| Scope          | Resource group     |

## 1. Objective

The objective of this lab was to use a Microsoft Entra ID security group and Azure RBAC to give a user read-only access to a resource group, instead of assigning the role to the user directly. The lab also checked that this read-only access did not allow the user to create new resources.

## 2. Environment

The lab used the following resources:

|                 |                         |
|-----------------|-------------------------|
| Resource group  | rg-gp-access-model      |
| Storage account | stgpaccessmodel65680763 |
| Security group  | gp-rg-readers65680763   |
| RBAC role       | Reader                  |
| Scope           | Resource group          |

Home > Resource Manager | Resource groups

### Create a resource group

Basics Tags Review + create

[Automation Link](#)

**Basics**

|                     |                              |
|---------------------|------------------------------|
| Subscription        | MOC Subscription-Iod53256297 |
| Resource group name | rg-gp-access-model           |
| Region              | North Europe                 |

**Tags**

None

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Figure 1 — Review + create screen for the resource group `rg-gp-access-model` in North Europe.

**Missing screenshot evidence.** No screenshot of the `stgpaccessmodel65680763` storage account being created was captured. Its name is confirmed later in Figure 6, where a second, unrelated storage account was blocked in the same resource group.

## 3. Identity and Group Setup

A security group, `gp-rg-readers65680763`, was created to hold the users who should have read-only access to the resource group. A pre-provided lab user, `Alexgp-65680763`, was then added to that group.

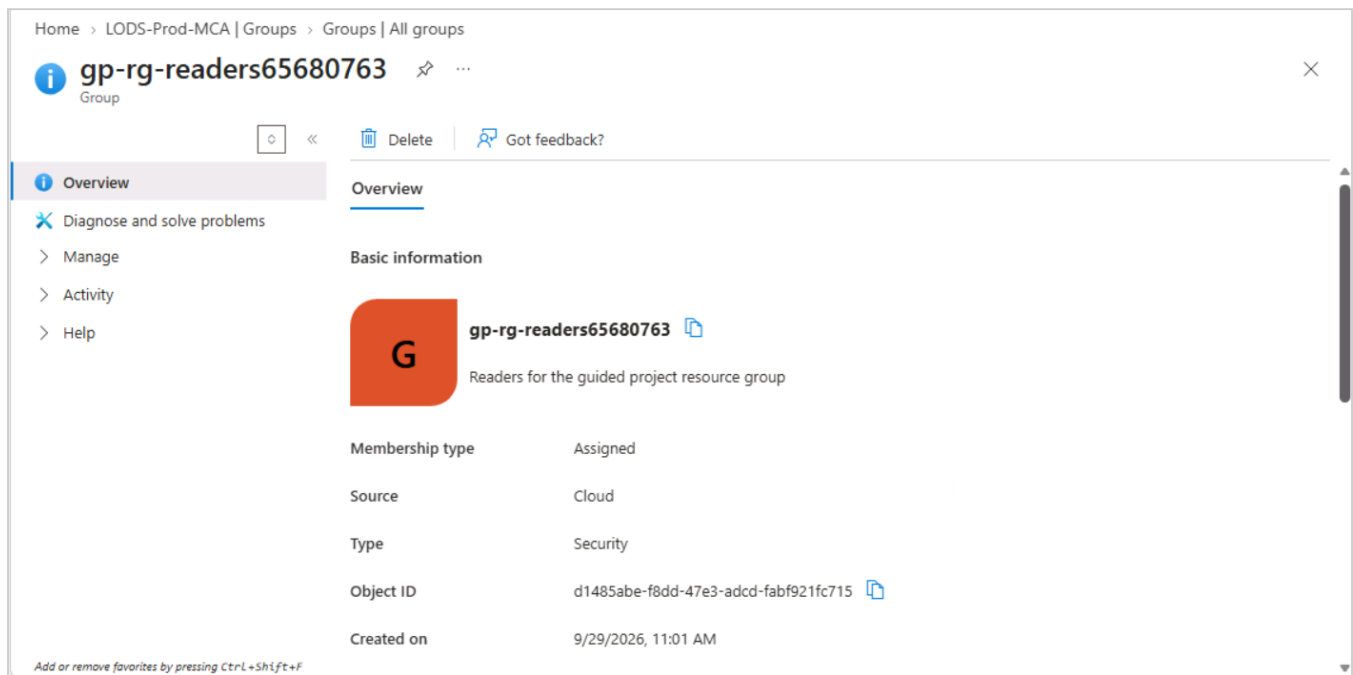


Figure 2 — Overview of the security group *gp-rg-readers65680763*, described as "Readers for the guided project resource group".

The portal confirmed the addition with a "Member successfully added" notification for Alexgp-65680763.

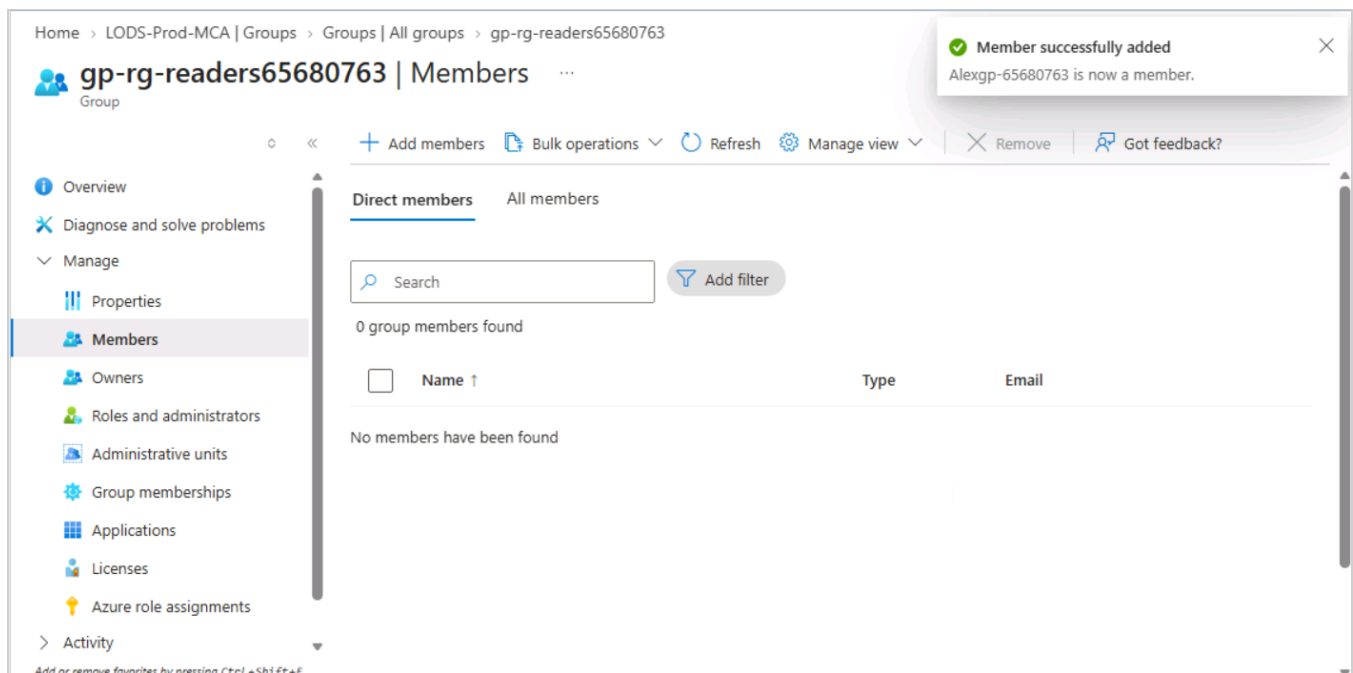


Figure 3 — Confirmation toast: "Alexgp-65680763 is now a member."

**Note:** the member list on this same screen still shows "0 group members found" — most likely the list had not refreshed yet when the screenshot was taken. The toast notification is the evidence used here for the membership having been added.

## 4. RBAC Assignment

The Reader role was assigned to the security group at the resource-group scope, rather than to Alexgp-65680763 directly. Assigning the role to the group means any user added to gp-rg-readers65680763 automatically inherits the same read-only access, without a separate role assignment being needed for each person.

Home > Resource Manager | Resource groups > rg-gp-access-model | Access control

### Add role assignment

Role **Members** Conditions Assignment type Review + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)  
[View my access](#)

**Selected role** Reader

**Assign access to** ☒ User, group, or service principal  
☐ Managed identity

**Members** + Select members

| Name                | Object ID |
|---------------------|-----------|
| No members selected |           |

**Description** Optional

[Review + assign](#) [Previous](#) [Next](#)

### Select members

gp-rg-readers65680763

gp-rg-readers65680763  
Readers for the guided project resource group

**Selected members:**

No members selected. Search for and add one or more members you want to assign to the role for this resource.

[Learn more about RBAC](#)

[Select](#) [Close](#)

Figure 4 — Add role assignment panel: role Reader, with the group gp-rg-readers65680763 being selected as the assignment target.

The resource group's Access control (IAM) page then showed the Reader role assigned to the group, scoped to "This resource".

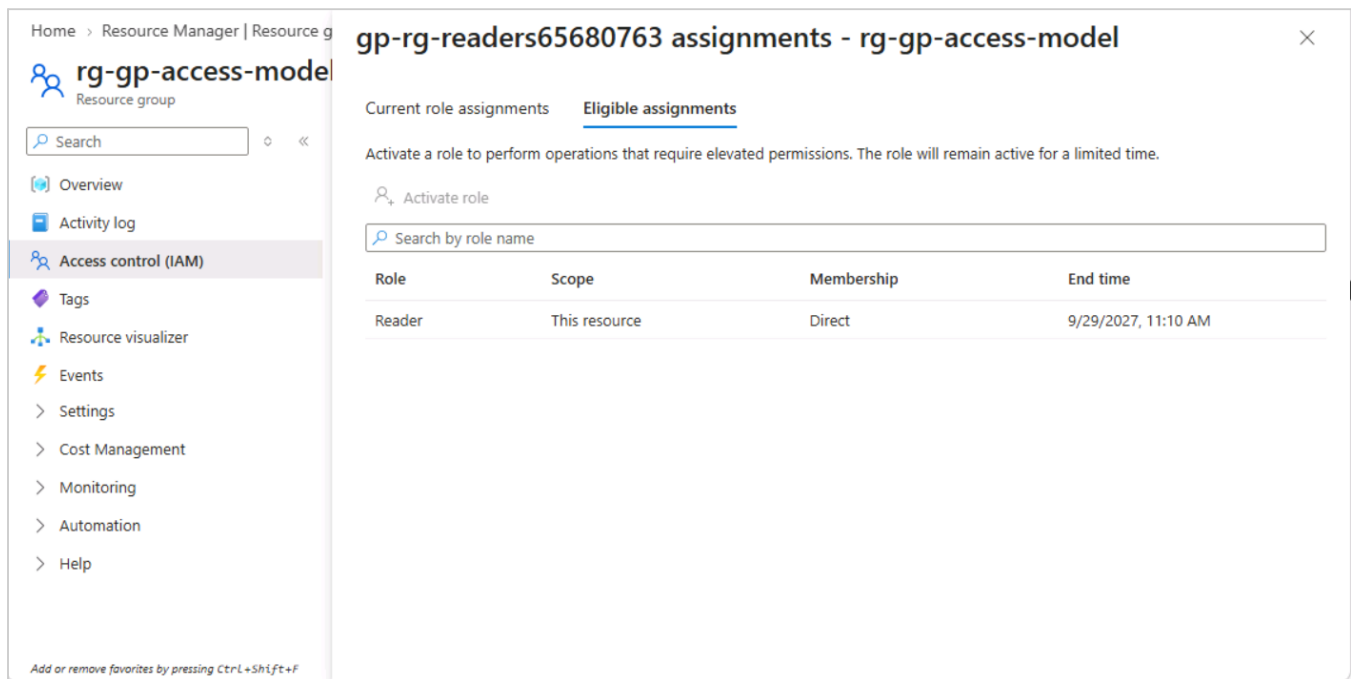


Figure 5 — gp-rg-readers65680763 assignments on rg-gp-access-model: Role Reader, Scope This resource, Membership Direct.

**Note:** this screenshot is on the "Eligible assignments" tab, not "Current role assignments". That means this lab environment used Privileged Identity Management (PIM), and the Reader role shown here is eligible to be activated rather than confirmed as permanently active. No screenshot of an active assignment was captured.

## 5. Access Verification

The lab steps included using the Check access tool on the resource group to confirm that Alexgp-65680763 inherited Reader access through the gp-rg-readers65680763 group.

**Missing screenshot evidence.** No screenshot of the Check access result was captured, so it is not included here.

## 6. Audit Trail

The lab steps also included reviewing the Azure Activity Log for the "Create role assignment" event, to confirm the role assignment was recorded.

**Missing screenshot evidence.** No screenshot of the Activity Log entry was captured, so it is not included here.

## 7. Least-Privilege Test

The lab called for signing in as Alexgp-65680763 to verify that the resource group and its resources could be viewed, then attempting to create a new storage account to confirm that Reader access does not allow resource creation.

**Missing screenshot evidence.** No screenshot of signing in as Alexgp-65680763 or viewing the resource group as that user was captured, so read access is not directly evidenced here beyond the role assignment shown in Section 4.

The write-access test is evidenced: while signed in as Alexgp-65680763, an attempt to create a new storage account (stgptestperm) in rg-gp-access-model was blocked with an AuthorizationFailed error, because the Reader role does not include permission to create resources.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user is signed in as 'Alexgp-65680763@LOD...'. The main content area is titled 'Create a storage account' and shows a 'Validation failed' message with a red 'X' icon. Below this, the 'Basics' tab is selected, showing details for the storage account 'stgptestperm' in the resource group 'rg-gp-access-model'. The error details pane on the right, titled 'Errors', shows the 'Summary' tab with the following message: 'The client 'Alexgp-65680763@LODSPRODMCA.onmicrosoft.com' with object id 'f0c79db6-7647-4e06-a94a-e5ca3259e409' does not have authorization to perform action 'Microsoft.Resources/deployments/validate/action' over scope '/subscriptions/40c4ed33-08cc-44ef-bc52-5c826af97410/resourceGroups/rg-gp-access-model/providers/Microsoft.Resources/deployments/stgptestperm\_1790706059935' or the scope is invalid. If access was recently granted, please refresh your credentials. (Code: AuthorizationFailed)'. Below the error message, there are links for 'Explain with Copilot', 'New Support Request', and 'Give Feedback'.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Storage center | Blob Storage

### Create a storage account

Validation failed. View error details →

Basics Advanced Networking Data protection Security

View automation template

**Basics**

|                      |                                               |
|----------------------|-----------------------------------------------|
| Subscription         | MOC Subscription-lod53256297                  |
| Resource group       | rg-gp-access-model                            |
| Location             | North Europe                                  |
| Storage account name | stgptestperm                                  |
| Primary service      | Azure Blob Storage or Azure Data Lake Storage |
| Performance          | Standard                                      |
| Replication          | Locally redundant storage (LRS)               |

**Advanced**

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### Errors

Summary Raw Error

Copy Full Error

**TOP ERROR**

The client 'Alexgp-65680763@LODSPRODMCA.onmicrosoft.com' with object id 'f0c79db6-7647-4e06-a94a-e5ca3259e409' does not have authorization to perform action 'Microsoft.Resources/deployments/validate/action' over scope '/subscriptions/40c4ed33-08cc-44ef-bc52-5c826af97410/resourceGroups/rg-gp-access-model/providers/Microsoft.Resources/deployments/stgptestperm\_1790706059935' or the scope is invalid. If access was recently granted, please refresh your credentials. (Code: AuthorizationFailed)

WAS THIS HELPFUL? 👍 🗑

Explain with Copilot

**Troubleshooting Options**

New Support Request

**Give Feedback**

Tell us about your experience with the Errors page

Figure 6 — Signed in as Alexgp-65680763, attempting to create the storage account stgptestperm — blocked with "AuthorizationFailed".

## 8. What I Learned

- How to create a Microsoft Entra ID security group and add a user to it
- How Azure RBAC roles can be assigned to a group instead of to individual users
- How to assign a role at the resource-group scope using Access control (IAM)
- That this lab environment used Privileged Identity Management (PIM) eligible assignments, and how that differs from an always-on active assignment
- That the Reader role allows viewing resources but does not allow creating or changing them
- How Azure reports a denied operation through an AuthorizationFailed error

## 9. Conclusion

This lab showed how assigning the Reader role to a security group, rather than to an individual user, can provide controlled access to a resource group: any user added to the group inherits the same permissions. The attempted storage account creation confirmed that Reader access does not allow resource changes. Some verification steps (Check access, Activity Log, and signing in to confirm read access) were part of the lab but were not captured in screenshots, so they are noted as gaps rather than claimed as proven here. This was a training environment, not a production implementation.